

Lab 12: Clustering

Learning Objectives

1. Clustering
 2. Unsupervised Learning
 3. K-Means Clustering
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1. Clustering

Cluster analysis is a branch of data science that deals with finding similarities between data according to the characteristics found in the data and grouping similar data objects into clusters. The term cluster means groups or partitions. In data science, **cluster is a collection of data objects** that are similar to one another within the same cluster and dissimilar to the objects in other clusters. A good clustering method will produce high quality clusters with – **high intra-class similarity** (Similar to one another within the same cluster) and **low inter-class similarity** (Dissimilar to the objects in other clusters).

Distances (usually Euclidean distance) are normally used to measure the similarity or dissimilarity between two data objects.

In clustering, we do not have a target to predict. We look at the data and then try to club similar observations and form different groups. Hence it is an unsupervised learning problem.

There are many clustering algorithms among them the most common and basic one is K-Means that we will cover in this lab.

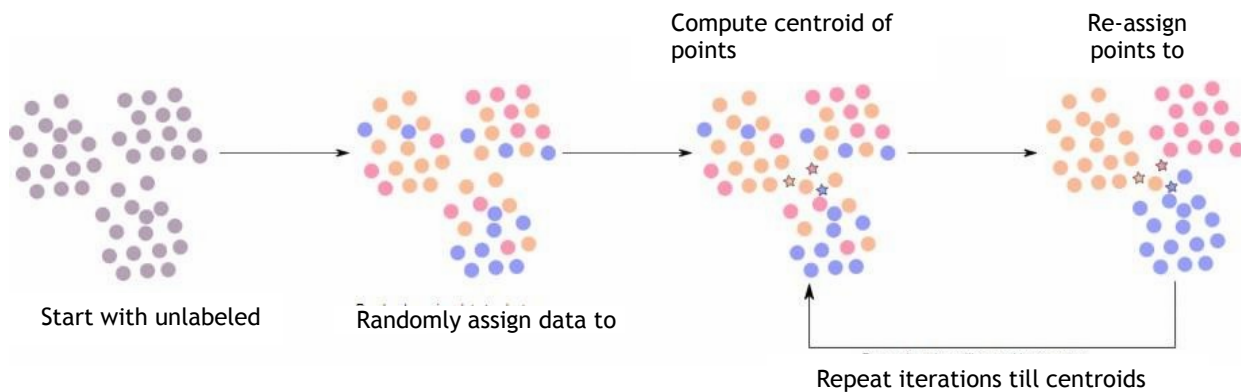
2. K-Means Clustering

K-means clustering is one of the simplest and popular unsupervised machine learning algorithms. Typically, unsupervised algorithms make inferences from datasets using only input vectors without referring to known, or labelled, outcomes.

K-means is a simple iterative clustering algorithm. Starting with randomly chosen K centroids, the algorithm proceeds to update the centroids and their clusters to equilibrium while minimizing the total within cluster variance. Its visual representation is given in below figure.

It is primarily used in scenarios with real-valued features because it relies on the Euclidean distance to discover cluster centroids.

The main objective of the K-Means algorithm is to minimize the sum of distances between the points and their respective cluster centroid.



Which Column is a Target Column?

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
140	141	Female	57	75	5
39	40	Female	20	37	75
45	46	Female	24	39	65
3	4	Female	23	16	77
199	200	Male	30	137	83
24	25	Female	54	28	14
152	153	Female	44	78	20
150	151	Male	43	78	17
168	169	Female	36	87	27
64	65	Male	63	48	51

2.1. K-MEANS ALGORITHM:

Given k, the k-means algorithm is implemented in four steps:

1. Partition objects into k nonempty subsets
2. Compute seed points as the centroids of the clusters of the current partition (the centroid is the center, i.e., mean point, of the cluster)

3. Assign each object to the cluster with the nearest seed point (use Euclidean distance to determine which data point is nearest to which cluster)
4. Go back to Step 2, stop when stopping criteria is reached.

2.2. STOPPING CRITERIA:

- The centroids have stabilized — there is no change in their values because the clustering has been successful.
- The defined number of iterations has been achieved.

2.3 Code:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

data = pd.read_csv("/content/customer-segmentation-clustering/customers.csv")

X = data[['Annual Income (k$)', 'Spending Score (1-100)']]

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# .fit
kmeans = KMeans(n_clusters=5, random_state=42)
data['Cluster'] = kmeans.fit_predict(X_scaled)

plt.figure()
plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=data['Cluster'])
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.title("Customer Segmentation using K-Means")
plt.show()
```

Lab Task

1. Consider the below notebook and implement K-Means on Mall Customer Segmentation Dataset. Also you are required to explore the following Notebook to understand how EDA is done and what insights are concluded from each Visualization.

<https://www.kaggle.com/code/kaushikp9699/customer-segmentation-clustering>

➔ Dataset is uploaded on Google Classroom.

Lab # 12 Marks distribution

	ER1	ER6	ER8
Task	3 points	3 points	4 points

Lab # 12 Rubric Evaluation Guideline:

#	Qualities & Criteria	0 < Poor <= 1	1 < Satisfactory <= 2	2 < Excellent <=3
ER1	Task Completion	Minimal or no program functionality was achieved.	Some tasks were completed, but the program has errors or incomplete functionalities.	All tasks were completed, and the program runs without errors.
#	Qualities & Criteria	0 < Poor <= 1	1 < Satisfactory <= 2	2 < Excellent <=3
ER6	Program Output	Output is inaccurate or poorly presented.	Output is mostly accurate but may lack labels, captions, or formatting.	Output is clear, accurate, and well presented with labels, captions, and proper formatting.
#	Qualities & Criteria	0 < Poor <= 1	1.5 < Satisfactory <= 2.5	2.5 < Excellent <= 4
ER8	Question & Answer	Answers some questions but not confidently or based on lab task knowledge.	Answers most questions confidently and based on lab task knowledge.	Answers all questions confidently and demonstrates a deep understanding of the given lab task.